



Fetal/Neonatal Morbidity and Mortality

Drugs that act directly on the renin-angiotensin system can cause fetal and neonatal morbidity and death when administered to pregnant women. In patients who were taking angiotensin converting enzyme inhibitors several dozen cases have been reported. When pregnancy is detected Losartan potassium should be discontinued as soon as possible.

The use of drugs that act directly on the renin-angiotensin system during the second and third trimesters of pregnancy has been associated with fetal and neonatal injury, including hypotension, neonatal skull hypoplasia, anuria, reversible or irreversible renal failure, and death. Oligohydramnios, presumably resulting from decreased fetal renal function has also been reported. Oligohydramnios in this setting has been associated with fetal limb contractures, craniofacial deformation, and hypoplastic lung development. Prematurity, intrauterine growth retardation, and patent ductus arterieus have also been reported, although it is not clear whether these occurrences were due to exposure to the drug.

These adverse effects do not appear to have resulted from intrauterine drug exposure that has been limited to the first trimester. Mothers whose embryos and fetuses are exposed to an angiotensin II receptor antagonist only during the first trimester should be so informed. Nonetheless, when patients become pregnant, physicians should have the patient discontinue the use of Losartan potassium as soon as possible. Rarely (probably less often than once in every thousand pregnancies), no alternative to an angiotensin II receptor antagonist will be found. In these rare cases, the mothers should be apprised of the potential hazards to their fetuses, and serial ultrasound examinations should be performed to assess the intra-amniotic environment.

Unless it is considered life-saving for the mother Losartan potassium should be discontinued if oligohydramnios is observed. Depending upon the week of pregnancy contraction stress testing (CST), a non-stress test (NST), or biophysical profiling (BPP) may be appropriate. However, patients and physicians should be aware, that oligohydramnios may not appear until after the fetus has sustained irreversible injury. Infants with histories of in utero exposure to an angiotensin II receptor antagonist should be closely observed for hypotension, oliguria, and hyperkalemia. If oliguria occurs, attention should be directed toward support of blood pressure and renal perfusion. As means of reversing hypotension and/or substituting for disordered renal function exchange, transfusion or dialysis may be required.

Hypotension and electrolyte/fluid imbalance

In patients who are intravascularly volume-depleted (e.g., those treated with diuretics) symptomatic hypotension may occur after initiation of therapy with Losartan potassium. These conditions should be corrected prior to administration of Losartan potassium, or a lower starting dose should be used.

In pediatric patients who are intravascularly volume-depleted, these conditions should be corrected prior to administration of Losartan.

Electrolyte imbalances are common in patients with renal impairment, with or without diabetes, and should be addressed.

Interaction with other medicaments

No significant drug-drug pharmacokinetic interactions have been found with hydrochlorothiazide, digoxin, warfarin, cimetidine and phenobarbital. Rifampin, an inducer of drug metabolism, decreased the concentrations of Losartan and its active metabolite. In human, two inhibitors of P450 3A4 have been studied. After intravenous administration of Losartan, ketoconazole did not affect the conversion of Losartan to the active metabolite, and erythromycin had no clinically significant effect after oral administration. Fluconazole, an inhibitor of P450 2C9, decreased active metabolite concentration and increased Losartan concentration. The pharmacodynamics consequences of concomitant use of Losartan and inhibitors of P450 2C9 have not been examined. Subjects who do not metabolize Losartan to active metabolite have been shown to have a specific, rare defect in cytochrome P450 2C9. These data suggest that the conversion of Losartan to its active metabolite is mediated primarily by P450 2C9 and not P450 3A4.

Concomitant use of potassium-sparing diuretics (e.g., spironolactone, triamterene, amiloride), potassium supplements, or salt substitutes containing potassium may lead to increases in serum potassium as with other drugs that block angiotensin II or its effects.

Lithium

As with other drugs which affect the excretion of sodium lithium's excretion may be reduced. Serum lithium levels should therefore be monitored carefully if lithium salts are to be co-administered with angiotensin II receptor antagonists.

Non-Steroidal Anti-Inflammatory Agents including Selective Cyclooxygenase-2 Inhibitors

In some patients with compromised renal function who are being treated with non-steroidal anti- inflammatory drugs (NSAID's) including those that selectively inhibit cyclooxygenase-2 inhibitors (COX-2 inhibitors). The co-administration of angiotensin II receptor antagonists including Losartan may result in a further deterioration of renal function .These effects are usually reversible. The antihypertensive effect of angiotensin II receptor antagonists, including Losartan may be diminished by NSAID's including selective COX-2 inhibitors. This interaction should be given consideration in patients taking NSAID's including selective COX-2 inhibitors concomitantly with angiotensin II receptor antagonists.

Pregnancy and Lactation

Pregnancy

Pregnancy Category C (first trimester) and D (second and third trimester)

In animals Losartan potassium has demonstrated foetal and neonatal injury and death, although there is no experience with the use of Losartan in pregnant women. The mechanism of which is believed to be pharmacologically mediated through effects on the renin-angiotensin-aldosterone system.

In humans, foetal renal perfusion, which is dependent upon the development of the renin- angiotensin-aldosterone system, begins in the second trimester; thus, if Losartan is administered during the second or third trimesters of pregnancy risk to the foetus increases.

Drugs that act directly on the renin-angiotensin-aldosterone system can cause injury and even death in the developing fetus when used during the second and third trimesters of pregnancy. Losartan should not be used in pregnancy, and if pregnancy is detected Losartan should be discontinued as soon as possible.

Lactation

Excretion of Losartan in human milk is not known, but significant levels of Losartan and its active metabolite were shown to be present in rat milk, because of the potential for adverse effects on the nursing infant, a decision should be made whether to discontinue breast-feeding or discontinue the drug, taking into account the importance of the drug to the mother.

Side Effects

In hypertensive patients with left ventricular hypertrophy Losartan was generally well tolerated. Dizziness, asthenia/fatigue and vertigo were the most common drug-related side effects. In type 2 diabetic patients with nephropathy Losartan was generally well tolerated. The most common drug-related side effects were asthenia/fatigue, dizziness, hypotension and hyperkalaemia., Due to hyperkalaemia few patients discontinued the study.

Symptom and treatment of overdose

Symptoms of intoxication

Limited data are available with regard to overdose in humans. The most likely manifestation of overdose would be hypotension and tachycardia. Bradycardia could occur from parasympathetic (vagal) stimulation.

Treatment of intoxication

If symptomatic hypotension should occur, supportive treatment should be instituted. Measures are depending on the time of medicinal product intake and kind and severity of symptoms. Stabilisation of the cardiovascular system should be given priority. After oral intake, the administration of a sufficient dose of activated charcoal is indicated. Afterwards, close monitoring of the vital parameters should be performed. Vital parameters should be corrected if necessary.

Neither Losartan nor the active metabolite can be removed by haemodialysis.

Storage

Store below 30 °C.

Shelf life

24 months

Presentation

LOSAGEN 25, LOSAGEN 50 and LOSAGEN 100 is proposed to be marketed in 10 x 10 tablets Blister pack and 3 x 10 tablets Blister pack. Losartan Potassium Tablets packed in blisters (Alu/PVC/PE/PVdC and Alu/PVC) will be further packed in pre-printed cartons with package leaflet according to the approved pack size.

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Date of Revision: 21/06/2024